

RBE 500 – Group Assignment – Part 2

This assignment is about moving the robot on a linear path.

- 1) (6 pts) Velocity Level Kinematics: Implement a node with two services. One takes joint velocities and converts them to end effector velocities, and the second one takes end effector velocities and converts them to joint velocities.
- 2) (3.5 pts) Write a node that provides incremental position references to the robot joints, i.e. $q_ref = q_ref_old + \Delta q * sampling_time$. The node would then send the q_ref to the joint position controllers of the robot as joint goals. You will use this node just like a velocity controller as follows in the next item.
- 3) (1.5) Give a constant velocity reference in the positive 'y' direction of the Cartesian space. Convert this velocity to the joint space velocities using your Jacobian and feed it as a reference (Δq) to your incremental position reference node. This should make the robot move on a straight line in the +y direction. Record the followed cartesian position from the output of your forward kinematic node, and plot it with respect to time (you can use easy plotting functions of Matlab). Record a video of the robot moving on a line and include in the report (please be sure to shrink the video size so that it does not create submission problems).

Write a report about your implementation. The report does not have to be long, but it should explain all the steps of the implementation. Copy-pasting the code and the results is not enough. Submit your report together your node.